Emerging Technologies in the Modern Era

DCIT 26 – Application Development and Emerging Technologies

Bachelor of Science in Computer Science

Submitted by:

Macatumpag, Elijah D.

Date:

September 12, 2026

Instructor:

Johneros Puyo

Overview of Emerging Technology

When the word “Emerging Technology” comes to mind, some people will think that it’s a technology that will soon “emerge,” per se, a new technology that will be of good use in the future, or a newly introduced invention that will help our lives get better. Everyone has a different perspective and opinion on these words, though these characteristics are not entirely accurate.

Sadiku et al. (2025) state that emerging technologies are technologies whose development, advancement, and practical applications are all still largely unrealized. This means they already exist but are still developing, evolving, and changing. What makes it “emerge” is that it has great potential to change the status quo.

To be clear, emerging technologies refer to technologies that are still progressing, evolving, and have the potential to change the current industry or may have a huge impact on our lives. We can label a technology as an emerging technology with the characteristics below.

Characteristics of Emerging Technologies (Rotolo et al.)

Characteristics	Definition
Radical Novelty	Creating something new or putting an existing technology to a new use. Generative AI is an example of radical novelty. It uses AI to create text, images, or code in a new way.
Relatively Fast Growth	The fast growth of a technology can be seen through the increased number of users, technological advancements, knowledge, and outputs produced.
Coherence	Refers to the collaboration of entirely different technologies or research streams to achieve a new purpose.
Prominent Impact	Changes made by a technology to different sectors such as business, education, and healthcare.
Uncertainty & Ambiguity	Many possible outcomes mean uncertainty. The future is unpredictable, and so are emerging technologies.

Emerging Technology Areas

- **Augmented Reality (AR)** – Merging the real world with the digital world. Pokémon Go is a great example of AR where you can view the Pokémon you caught through a camera and take pictures with it. AR is also used in furniture displays. For instance, an app lets you see a piece of furniture through a phone first before buying it. AR helps people immerse themselves in the digital world to solve real-life problems.

- **Artificial Intelligence (AI)** – Simulating human intelligence in a machine. Much like a thinking human, AI was invented to perform tasks that normally require human thinking, such as learning, reasoning, and correcting mistakes.
    - **Machine Learning** – Providing a machine with the ability to learn from experience, much like a human, is what makes machine learning possible. It is widely used in various applications such as healthcare, transportation, marketing, and financial services.
- **Quantum Computers** - A normal computer excels at everyday sequential tasks like web browsing, emails, gaming, or office applications, but quantum computers use quantum mechanics to perform calculations that can solve some problems far more efficiently than a normal computer.

The Fourth Industrial Revolution

Technology advances as well as its production. The stages of production technology are called the "industrial revolution." It refers to the systematic transformation of how technologies are manufactured through the use of physical and digital systems to enhance quality, economy, and efficiency. These advancements change the environment and working conditions of the people. There are four (4) stages of the Industrial Revolution:

- **First Industrial Revolution (Industry 1.0)** – Began in the 18th century. Introduced the production of mechanical components using hydroelectric and steam-powered equipment.
- **Second Industrial Revolution (Industry 2.0)** – Began in the 19th century. More enhanced and polished production was made with the use of electricity.
- **Third Industrial Revolution (Industry 3.0)** – The era where first computers were introduced. With the introduction of internet access and connectivity, automation and production processes were further enhanced.
- **Fourth Industrial Revolution (Industry 4.0)** – The current developing state. Cutting-edge technologies such as the Internet of Things (IoT), robotics, virtual reality (VR), AR, and AI are shaping our daily lives.

The Building Blocks of the Fourth Industrial Revolution

Big Data Analytics	Assists in making faster, data-based solutions in production through the process of collecting, organizing, and analyzing large sets of data from different resources.

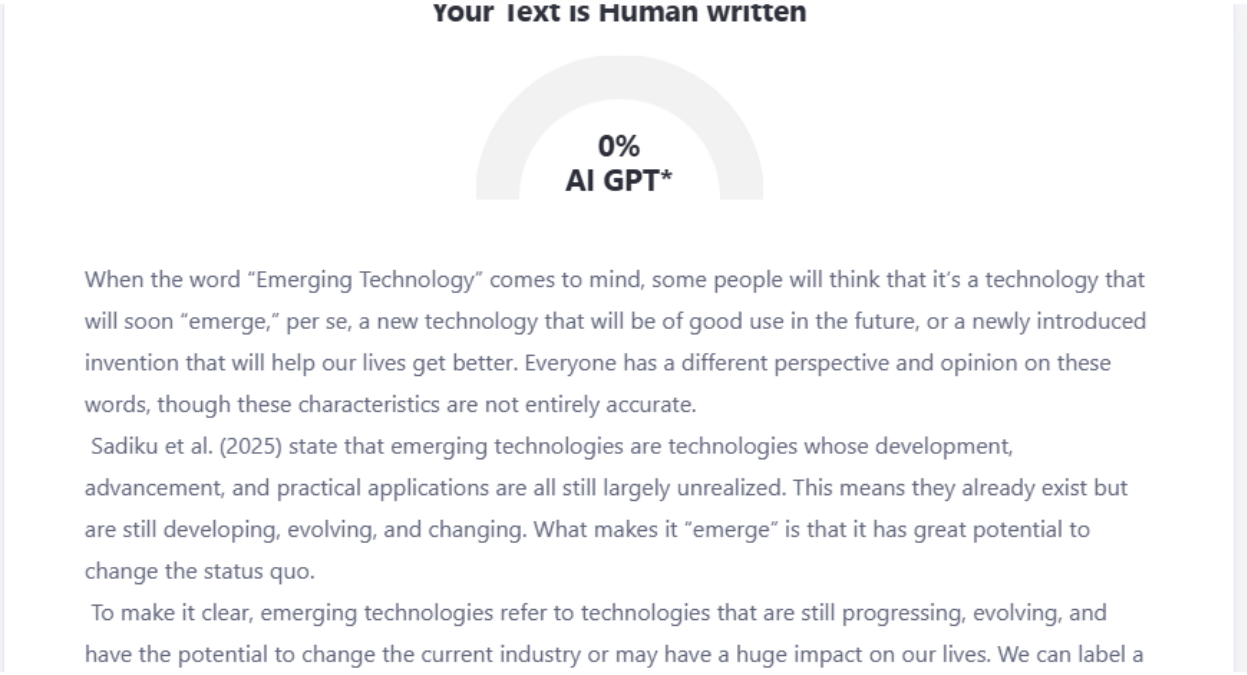
Internet of Things (IoT)	These are the physical components that can connect to the internet. Through the use of the internet, they can communicate with other devices.
Cloud Computing	The use of online servers and services through the Internet to store data, run applications, and access computing resources. It allows organizations to process and manage large amounts of data without relying entirely on local computers.
Augmented Reality	Immersing the digital world in the real world. In workplaces, AR can provide workers with real-time instructions, such as showing repair procedures or helping them identify parts in a warehouse.

References

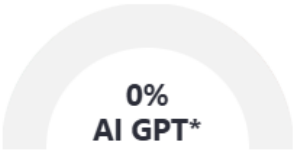
Sadiku et al. (2025). Emerging Technologies: Overview. ResearchGate.net. https://www.researchgate.net/publication/396653452_Emerging_Technologies_Overview

Rotolo et al. (2015). What is emerging technology?. ScienceDirect.com. https://www.sciencedirect.com/science/article/abs/pii/S0048733315001031

AI Checker and Grammarly Results

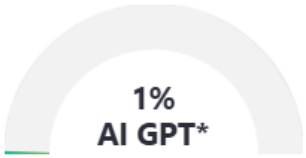


Your Text is Human written



• Augmented Reality (AR) – Merging the real world with the digital world. Pokémon Go is a great example of AR where you can view the Pokémon you caught through a camera and take pictures with it. AR is also used in furniture displays. For instance, an app lets you see a piece of furniture through a phone first before buying it. AR helps people immerse themselves in the digital world to solve real-life problems.
• Artificial Intelligence (AI) – Simulating human intelligence in a machine. Much like a thinking human, AI was invented to perform tasks that normally require human thinking, such as learning, reasoning, and correcting mistakes.
o Machine Learning – Providing a machine with the ability to learn from experience, much like a human, is what makes machine learning possible. It is widely used in various applications such as healthcare, transportation, marketing, and financial services.
• Quantum Computers - A normal computer excels at everyday sequential tasks like web browsing, emails, gaming, or office applications, but quantum computers use quantum mechanics to perform calculations that can solve some problems far more efficiently than a normal computer.

Your File Content is Human written



Emerging Technologies in the Modern Era
DCIT 26 – Application Development and Emerging Technologies

Your Text is Human written



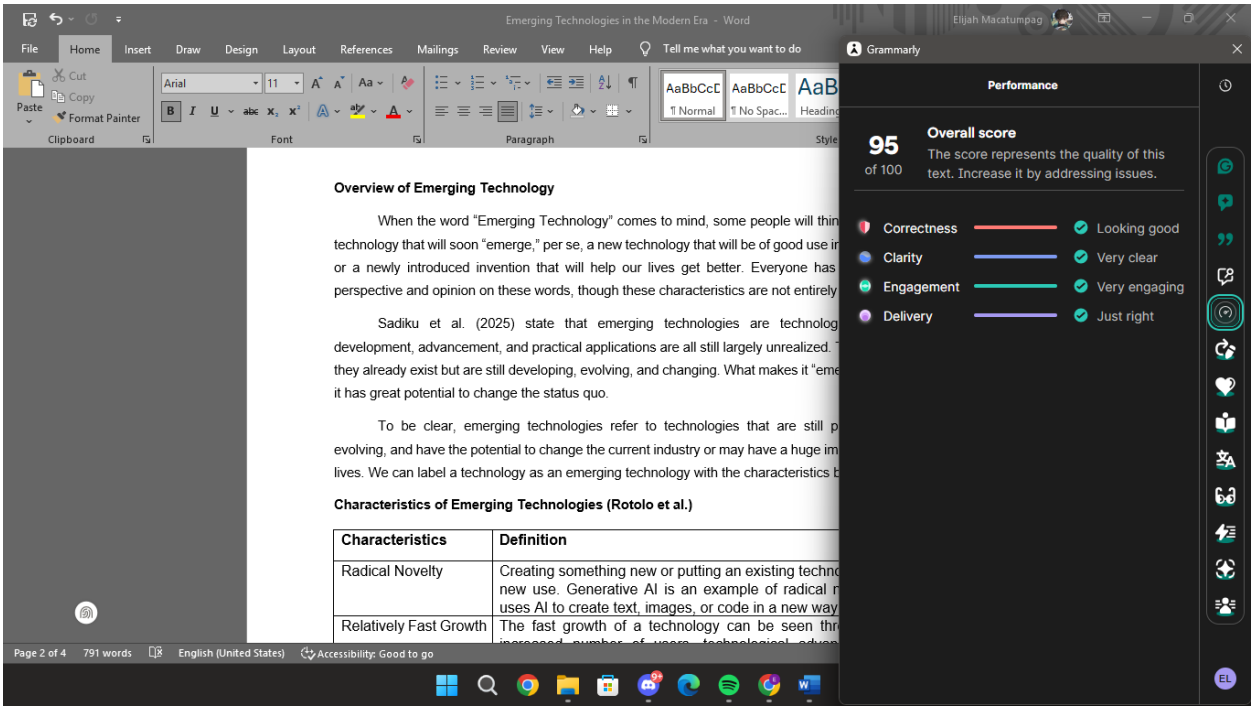
The Fourth Industrial Revolution

Technology advances as well as its production. The stages of production technology are called the "industrial revolution." It refers to the systematic transformation of how technologies are manufactured through the use of physical and digital systems to enhance quality, economy, and efficiency. These advancements change the environment and working conditions of the people. There are four (4) stages of the Industrial Revolution:

• First Industrial Revolution (Industry 1.0) – Began in the 18th century. Introduced the production of mechanical components using hydroelectric and steam-powered equipment.

• Second Industrial Revolution (Industry 2.0) – Began in the 19th century. More enhanced and polished production was made with the use of electricity.

• Third Industrial Revolution (Industry 3.0) – The era where first computers were introduced. With the introduction of internet access and connectivity, automation and production processes were further enhanced.



Criteria

Rubric:

Criteria	Weight
Accuracy of content (based on lecture)	
Completeness of all sections	
Depth of analysis and explanations	
Use of examples and applications	
Organization	
Originality (own words no copying)	

Total: 100%